

AI Career Intelligence Platform

Security Design Document

Threat Model • Security Controls • IAM • Encryption

Version: 1.0

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1. Threat Model

The platform protects user accounts, resumes, AI conversations, and job recommendation data from unauthorized access and misuse.

Threat	Mitigation
Unauthorized Login	JWT + bcrypt
SQL Injection	Parameterized queries / ORM
XSS	Input validation and output encoding
Sensitive Data Exposure	HTTPS and encryption
API Abuse	Rate limiting and authentication

2. Security Controls

- HTTPS for all communications.
- Server-side input validation.
- Role-based authorization.
- Audit logging for critical actions.
- Secure file upload validation.
- Dependency and vulnerability scanning.

3. Identity & Access Management (IAM)

Role	Permissions
User	Manage profile, resumes and AI features
Administrator	Manage users, jobs and system settings
System Service	Internal service-to-service access

Authentication uses JWT access tokens with hashed passwords (bcrypt). Future versions may support OAuth providers.

4. Encryption

Area	Approach
Data in Transit	HTTPS / TLS
Passwords	bcrypt hashing
JWT	Signed access tokens
Database Backups	Encrypted storage
Secrets	Environment variables / Secret Manager

Encryption protects user credentials, uploaded resumes and sensitive application data throughout the system lifecycle.

5. Conclusion

This document defines the security baseline for Version 1.0 of the AI Career Intelligence Platform, including threat mitigation, access control and encryption practices to support a secure production deployment.